Data Aggregation

Analyzing Sales Performance by Region in a Retail Company

Problem Statement: Analyze sales performance across locations and product categories to identify the top-performing locations.

Dataset: `Retail_Sales_Data.csv`

Task 1: Import the "Retail_Sales_Data.csv" dataset.

In [1]:
```
import pandas as pd
import matplotlib.pyplot as plt

df = pd.read_csv("Retail_Sales_data.csv")
display(df.head())
```

	invoice_no	customer_id	gender	age	category	quantity	price	payment_method	invoice_date	shopping_mall
0	I138884	C241288	Female	28	Clothing	5	1500.40	Credit Card	5/8/2022	Kanyon
1	I317333	C111565	Male	21	Shoes	3	1800.51	Debit Card	12/12/2021	Forum Istanbul
2	I127801	C266599	Male	20	Clothing	1	300.08	Cash	9/11/2021	Metrocity
3	I173702	C988172	Female	66	Shoes	5	3000.85	Credit Card	16/05/2021	Metropol AVM
4	I337046	C189076	Female	53	Books	4	60.60	Cash	24/10/2021	Kanyon

Task 2: Explore the dataset to understand its structure and content.

In [2]:
```
print("Dataset size:", df.shape)
print("\nColumn names:")
print(df.columns.tolist())

print("\nData types and non-null values:")
df.info()

print("\nMissing values:")
display(df.isnull().sum().to_frame("Missing Values"))

print("\nSummary of numeric columns:")
display(df.describe())
```

```
Dataset size: (99457, 10)

Column names:
['invoice_no', 'customer_id', 'gender', 'age', 'category', 'quantity', 'price', 'payment_method', 'invoice_date', 'shopping_mall']

Data types and non-null values:
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 99457 entries, 0 to 99456
Data columns (total 10 columns):
 #   Column          Non-Null Count  Dtype  
---  ------          --------------  -----  
 0   invoice_no      99457 non-null  object 
 1   customer_id     99457 non-null  object 
 2   gender          99457 non-null  object 
 3   age             99457 non-null  int64  
 4   category        99457 non-null  object 
 5   quantity        99457 non-null  int64  
 6   price           99457 non-null  float64
 7   payment_method  99457 non-null  object 
 8   invoice_date    99457 non-null  object 
 9   shopping_mall   99457 non-null  object 
dtypes: float64(1), int64(2), object(7)
memory usage: 7.6+ MB

Missing values:
```

	Missing Values
invoice_no	0
customer_id	0
gender	0
age	0
category	0
quantity	0
price	0
payment_method	0
invoice_date	0
shopping_mall	0

```
Summary of numeric columns:
```

	age	quantity	price
count	99457.000000	99457.000000	99457.000000
mean	43.427089	3.003429	689.256321
std	14.990054	1.413025	941.184567
min	18.000000	1.000000	5.230000
25%	30.000000	2.000000	45.450000
50%	43.000000	3.000000	203.300000
75%	56.000000	4.000000	1200.320000
max	69.000000	5.000000	5250.000000

Task 3: Identify the relevant variables for aggregating sales data, such as region, sales amount, and product category.

In [3]:
```
print("Location column available: shopping_mall")
print("Product category column: category")
print("Quantity column: quantity")
print("Unit price column: price")

# The dataset has no region or total sales column, so use shopping_mall as the location
# and calculate each transaction's sales amount from quantity multiplied by price.
df["sales_amount"] = df["quantity"] * df["price"]

display(df[["shopping_mall", "category", "quantity", "price", "sales_amount"]].head())
```

```
Location column available: shopping_mall
Product category column: category
Quantity column: quantity
Unit price column: price
```

	shopping_mall	category	quantity	price	sales_amount
0	Kanyon	Clothing	5	1500.40	7502.00
1	Forum Istanbul	Shoes	3	1800.51	5401.53
2	Metrocity	Clothing	1	300.08	300.08
3	Metropol AVM	Shoes	5	3000.85	15004.25
4	Kanyon	Books	4	60.60	242.40

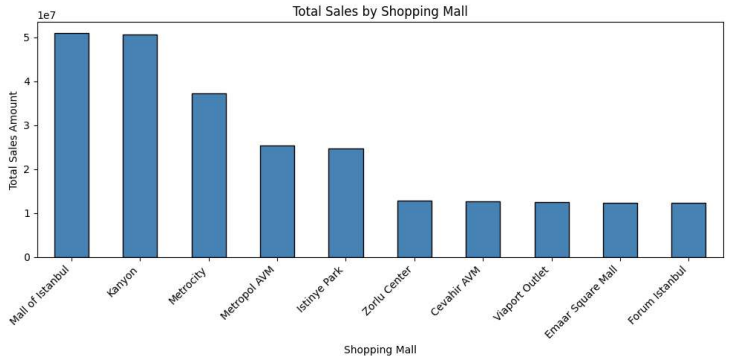
Task 4: Group the sales data by region and calculate the total sales amount for each region.

In [4]:
```
# Add up sales for each shopping mall.
sales_by_region = df.groupby("shopping_mall")["sales_amount"].sum().sort_values(ascending=False)
display(sales_by_region.to_frame("Total Sales Amount"))
```

shopping_mall	Total Sales Amount
Mall of Istanbul	50872481.68
Kanyon	50554231.10
Metrocity	37302787.33
Metropol AVM	25379913.19
Istinye Park	24618827.68
Zorlu Center	12901053.82
Cevahir AVM	12645138.20
Viaport Outlet	12521339.72
Emaar Square Mall	12406100.29
Forum Istanbul	12303921.24

Task 5: Create bar plots or pie charts to visualize the sales distribution by region.

In [5]:
```
plt.figure(figsize=(10, 5))
sales_by_region.plot(kind="bar", color="steelblue", edgecolor="black")
plt.title("Total Sales by Shopping Mall")
plt.xlabel("Shopping Mall")
plt.ylabel("Total Sales Amount")
plt.xticks(rotation=45, ha="right")
plt.tight_layout()
plt.show()
```



Task 6: Identify the top-performing regions based on the highest sales amount.

In [6]:
```
top_regions = sales_by_region.head(5)
print("Top-performing shopping malls by total sales:")
display(top_regions.to_frame("Total Sales Amount"))
```

Top-performing shopping malls by total sales:

shopping_mall	Total Sales Amount
Mall of Istanbul	50872481.68
Kanyon	50554231.10
Metrocity	37302787.33
Metropol AVM	25379913.19
Istinye Park	24618827.68

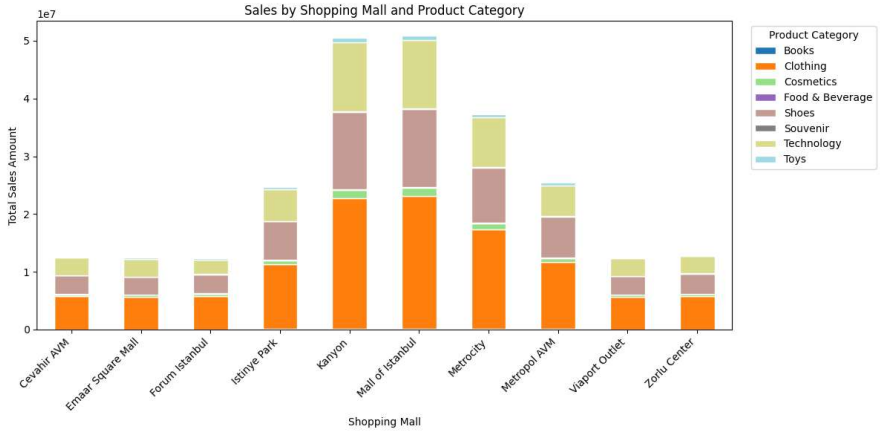
Task 7: Group the sales data by region and product category to calculate the total sales amount for each combination.

In [7]:
```
# Group sales by shopping mall and product category.
sales_by_region_category = df.pivot_table(
    index="shopping_mall",
    columns="category",
    values="sales_amount",
    aggfunc="sum",
    fill_value=0
)
display(sales_by_region_category)
```

shopping_mall	Books	Clothing	Cosmetics	Food & Beverage	Shoes	Souvenir	Technology	Toys
Cevahir AVM	44541.00	5706321.28	321214.00	44010.45	3243918.85	29723.82	3051300.0	204108.80
Emaar Square Mall	41995.80	5590490.40	338941.76	40610.95	3089675.16	30943.74	3094350.0	179092.48
Forum Istanbul	42056.40	5792444.24	353172.76	39162.24	3327942.65	32879.19	2516850.0	199413.76
Istinye Park	76083.30	11253900.24	655357.88	85918.44	6641481.22	68925.48	5436900.0	400261.12
Kanyon	163029.15	22609527.60	1369550.78	166497.05	13383190.83	127399.53	11944800.0	790236.16
Mall of Istanbul	172240.35	22947417.68	1367517.78	171177.90	13467814.80	127540.29	11828250.0	790522.88
Metrocity	125911.65	17226692.56	991860.04	129902.74	9519296.37	94227.09	8608950.0	605946.88
Metropol AVM	83718.90	11568084.00	680770.38	88638.04	7149825.21	67869.78	5327700.0	413306.88
Viaport Outlet	39632.40	5604594.16	347439.70	41662.18	3194704.91	27319.17	3066000.0	199987.20
Zorlu Center	45343.95	5697318.88	367037.82	41955.06	3535601.47	28996.56	2987250.0	197550.08

Task 8: Create stacked bar plots or grouped bar plots to compare the sales amounts across different regions and product categories.

In [8]:
```
sales_by_region_category.plot(
    kind="bar",
    stacked=True,
    figsize=(12, 6),
    colormap="tab20",
    edgecolor="white"
)
plt.title("Sales by Shopping Mall and Product Category")
plt.xlabel("Shopping Mall")
plt.ylabel("Total Sales Amount")
plt.legend(title="Product Category", bbox_to_anchor=(1.02, 1), loc="upper left")
plt.xticks(rotation=45, ha="right")
plt.tight_layout()
plt.show()
```


Sales by Shopping Mall and Product Category
1e7
Total Sales Amount
Shopping Mall
Cevahir AVM
Emaar Square Mall
Forum Istanbul
Istinye Park
Kanyon
Mall of Istanbul
Metrocity
Metropol AVM
Viaport Outlet
Zorlu Center
Product Category
Books
Clothing
Cosmetics
Food & Beverage
Shoes
Souvenir
Technology
Toys